Data Transformation and Feature Engineering

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

# Sample dataset
data = {
    'Area': [1000, 1500, 2000, 2500, 3000],
    'Bedrooms': [2, 3, 3, 4, 5],
    'Bathrooms': [1, 2, 2, 3, 3],
    'Price': [200000, 300000, 400000, 550000, 700000]
}

# Create DataFrame
df = pd.DataFrame(data)

print("Original Dataset:")
print(df)

# ----------------------------------
# a) DATA TRANSFORMATION
# ----------------------------------

# Log transformation of House Price
df['Log_Price'] = np.log(df['Price'])

# Scaling Area and Price using StandardScaler
scaler = StandardScaler()

df[['Scaled_Area', 'Scaled_Price']] = scaler.fit_transform(
    df[['Area', 'Price']]
)

# ----------------------------------
# b) FEATURE ENGINEERING
# ----------------------------------

# Create a new feature by combining
# Bedrooms and Bathrooms
```

```
df['Total_Rooms'] = df['Bedrooms'] + df['Bathrooms']

# Create another useful feature
# Area available per bedroom
df['Area_Per_Bedroom'] = df['Area'] / df['Bedrooms']

print("\nTransformed and Engineered Dataset:")
print(df)
```

```
Original Dataset:
   Area  Bedrooms  Bathrooms   Price
0  1000         2          1  200000
1  1500         3          2  300000
2  2000         3          2  400000
3  2500         4          3  550000
4  3000         5          3  700000

Transformed and Engineered Dataset:
   Area  Bedrooms  Bathrooms   Price  Log_Price  Scaled_Area  Scaled_Price  \
0  1000         2          1  200000  12.206073    -1.414214     -1.293851
1  1500         3          2  300000  12.611538    -0.707107     -0.731307
2  2000         3          2  400000  12.899220     0.000000     -0.168763
3  2500         4          3  550000  13.217674     0.707107      0.675053
4  3000         5          3  700000  13.458836     1.414214      1.518869

   Total_Rooms  Area_Per_Bedroom
0            3        500.000000
1            5        500.000000
2            5        666.666667
3            7        625.000000
4            8        600.000000
```